+32 494226259
maxime.lucas@unamur.be
maximelucas.github.io
maximelucas
ORCID ID: 0000-0001-8087-2981
date of birth: 25 Jan 1992
nationality: Belgian
Last updated: November 1, 2026

Maxime Lucas

Current position

Aug 26–now **Assistant Professor (chargé de cours)**, *UNamur*, Belgium
Mathematics Department and Namur Institute for Complex Systems (naXys)

Past positions

Apr 24–Aug 26 **FNRS postdoctoral fellow (CR)**, *UCLouvain and UNamur*, Belgium
Characterizing underground fungal networks for my project UNDER-NET funded by the FNRS, hosted by T. Carletti (naXys, UNamur) and S. Declerck (Team Mycology, UCLouvain).

Jun 22–Mar 24 **Postdoctoral researcher**, *CENTAI Institute*, Turin, Italy
Mainly higher-order interactions (synchronisation, structure), TDA applied to neuroscience, and deciphering sperm whale language (Project CETI), with G. Petri

Oct 21–May 22 **Postdoctoral researcher**, *ISI Foundation*, Turin, Italy
Mainly higher-order interactions (synchronisation, structure), TDA applied to neuroscience, and deciphering sperm whale language (Project CETI), with G. Petri

Jun 19–Aug 21 **CENTURI postdoctoral fellow**, *Aix-Marseille University*, Marseille, France
Cell cycle modelling as a temporal network of protein interactions with A. Barrat, B. Habermann, and L. Tichit.

Education

2015–2019 **PhD in Physics**, *Lancaster University (UK) and University of Florence (Italy)*
"Synchronization of oscillators with time-varying parameters", supervised by A. Stefanovksa (Lancaster) and D. Fanelli (Florence), funded by the Marie Curie ITN COSMOS.

2014–2015 **Master of Artificial Intelligence**, *Katholieke Universiteit Leuven*, Leuven, Belgium, *cum laude*
Thesis: Instabilities in Cortical Networks with Embedded Synfire Chains

2012–2014 **MSc Physics**, *Université Libre de Bruxelles*, Brussels, Belgium, *magna cum laude*
Thesis: Instabilités dynamiques de systèmes de billards avec interactions rares

2012 **One-term exchange programme**, *University of Toronto*, Toronto, Canada

2009–2012 **BSc Physics**, *Université Libre de Bruxelles*, Brussels, Belgium, *cum laude*

Publications

'*' indicates co-first authorship.

Manuscripts submitted

- Léon, I., Muolo, R., Zhang, Y. & **Lucas, M.** *Symmetry-based selection rules for higher-order interactions in coupled oscillators* under review at Physical Review Letters. 2026
- **Lucas*, M**, Bisot*, C., Petri, G., Declerck, S. & Carletti, T. *Minimal branching and fusion morphogenesis approaches biological multi-objective optimality* 2026
- Zaffina, L., Monti, C., Poetto, S., **Lucas, M.**, Machado Borges, H. J., Deshpande, R., Tønnesen, P., Gruber, D. F., Gero, S., Santoro, A., *et al. Sperm whale foraging combines persistent search with flexible prey-capture tactics* 2026
- Fujii, N., Moriamé, M., **Lucas, M.**, Nakao, H. & Carletti, T. *Higher-order interactions for controlling time-delayed Kuramoto model* 2026
- Skardal, P. S., Battiston, F., **Lucas, M.**, Mizuhara, M. S., Petri, G. & Zhang, Y. *Mixed higher-order coupling stabilizes new states* 2025

Review articles

- Battiston, F., Bick, C., **Lucas, M.**, Millán, A. P., Skardal, P. S. & Zhang, Y. Collective dynamics on higher-order networks. *Nature Reviews Physics* (2026).
- Battiston, F., Cencetti, G., Iacopini, I., Latora, V., **Lucas, M.**, Patania, A., Young, J.-G. & Petri, G. Networks beyond pairwise interactions: structure and dynamics. *Physics Reports* **874.** 1–92 (2020).

Research articles

- Maalouf*, A., DelPreto*, J., **Lucas*, M**, Poetto*, S., Andreas, J., Torralba, A., Gero, S., Petri, G., Rus, D. & Gruber, D. F. Cooperation by non-kin during birth underpins sperm whale social complexity. *Science* **391.** 1355–1360 (2026).
- Aluma, Y. *et al.* Description of a Collaborative Sperm Whale Birth and Shifts in Coda Vocal Styles during Key Events. *Scientific Reports* **16.** 9206 (2026).
- Moriamé, M., Muolo, R., Carletti, T. & **Lucas, M.** On the efficiency of pairwise Hamiltonian control to desynchronize the higher-order Kuramoto model. *Chaos* **36** (2026).
- **Lucas, M.**, Gallo, L., Ghavasieh, A., Battiston, F. & De Domenico, M. Reducibility of higher-order networks from dynamics. *Nature Communications* **17.** 1551 (2026).
- **Lucas, M.**, Francois, D., Mombaerts, L., Donato, C., Skupin, A. & Proverbio, D. Topological analysis of brain dynamical signals indicates signatures of seizure susceptibility. *Physical Review Research* (2025).
- Robiglio, T., Neri, M., Coppes, D., Agostinelli, C., Battiston, F., **Lucas, M.** & Petri, G. Synergistic signatures of group mechanisms in higher-order systems. *Physical Review Letters* **134.** 137401 (2025). ★ *Editors's suggestion & Featured in Physics.*
- Nurisso, M., Morandini, M., **Lucas, M.**, Vaccarino, F., Gili, T. & Petri, G. Higher-order Laplacian Renormalization. *Nature Physics* **21.** 661–668 (2025).

- **Lucas*, M**, Aime*, N., Callara, A., Fontanelli, L., Sebastiani, L., Santarcangelo, L. & Petri, G. On the state-nonstate theory of hypnosis: Network and topological EEG findings. *Cerebral Cortex* **35.** bhaf262 (2025).

- Moriamé, M., **Lucas, M.** & Carletti, T. Hamiltonian control to desynchronize Kuramoto oscillators with higher-order interactions. *Physical Review E* **111.** 044307 (2025).

- Brondetta, A., Bizyaeva, A., **Lucas, M.**, Petri, G. & Musslick, S. On the Benefits of Heterogeneity in Cognitive Stability and Flexibility for Collaborative Task Switching. *Proceedings of the Annual Meeting of the Cognitive Science Society* **46** (2024).

- Nurisso, M., Arnaudon, A., **Lucas, M.**, Peach, R. L., Expert, P., Vaccarino, F. & Petri, G. A unified framework for Simplicial Kuramoto models. *Chaos* **34.** 053118 (2024).

- Leitão, A., **Lucas, M.**, Poetto, S., Hersh, T. A., Gero, S., Gruber, D., Bronstein, M. & Petri, G. Evidence of social learning across symbolic cultural barriers in sperm whales. *eLife* **13** (2024).

- Santoro, A., Battiston, F., **Lucas, M.**, Petri, G. & Amico, E. Higher-order connectomics of human brain function reveals local topological signatures of task decoding, individual identification, and behavior. *Nature Communications* **15.** 10244 (2024).

- Zhang, Y., Skardal, P. S., Battiston, F., Petri, G. & **Lucas, M.** Deeper but smaller: Higher-order interactions increase linear stability but shrink basins. *Science Advances* **10.** eado8049 (2024).

- Zhang*, Y., **Lucas*, M** & Battiston, F. Higher-order interactions shape collective dynamics differently in hypergraphs and simplicial complexes. *Nature Communications* **14.** 1605 (2023).

- **Lucas, M.**, Townsend-Teague, A., Neri, M., Poetto, S., Morris, A., Habermann, B. H. & Tichit, L. Phasik: a Python package to identify system states in partially temporal networks. *Journal of Open Source Software* **8.** 5872 (2023).

- Landry, N. W., **Lucas, M.**, Iacopini, I., Petri, G., Schwarze, A., Patania, A. & Torres, L. XGI: A Python package for higher-order interaction networks. *Journal of Open Source Software* **8.** 5162 (2023).

- **Lucas*, M**., Iacopini*, I., Robiglio, T., Barrat, A. & Petri, G. Simplicially driven simple contagion. *Physical Review Research* **5.** 013201 (2023).

- **Lucas, M.**, Morris, A., Townsend-Teague, A., Tichit, L., Habermann, B. H. & Barrat, A. Inferring cell cycle phases from a partially temporal network of protein interactions. *Cell Reports Methods*. 100397 (2023).

- Newman, J., **Lucas, M.** & Stefanovska, A. Stabilization of cyclic processes by slowly varying forcing. *Chaos* **31.** 123129 (2021).

- **Lucas, M.**, Cencetti, G. & Battiston, F. Multiorder Laplacian for synchronization in higher-order networks. *Physical Review Research* **2.** 033410 (2020).

- **Lucas, M.**, Fanelli, D. & Stefanovska, A. Nonautonomous driving induces stability in network of identical oscillators. *Physical Review E* **99.** 012309 (2019).

- **Lucas, M.**, Fanelli, D., Carletti, T. & Petit, J. Desynchronization induced by time-varying network. *Europhys. Lett.* **121.** 50008 (2018).
- **Lucas, M.**, Newman, J. & Stefanovska, A. Stabilization of dynamics of oscillatory systems by nonautonomous perturbation. *Physical Review E* **97.** 042209 (2018).
- Duncan, R. & **Lucas, M.** Verifying the Steane code with Quantomatic. *Electronic Proceedings in Theoretical Computer Science* **171.** 33–49 (2014).

Book chapters

- **Lucas, M.**, Cencetti, G. & Battiston, F. *Higher-Order Systems* *233* (Springer Nature, 2022).
- **Lucas, M.**, Newman, J. & Stefanovska, A. *Physics of Biological Oscillators* *85–110* (Springer, 2021).
- Newman, J., **Lucas, M.** & Stefanovska, A. *Physics of Biological Oscillators* *111–129* (Springer, 2021).

Presentations

Invited talks (14)

2026 **Networks, Dynamics, and Neuroscience**, Amsterdam, The Netherlands, 13th February
Title: Sync with higher-order interactions: basins and coupling functions

2025 **Higher-Order Interactions and complex systems: a joint BeyondTheEdge and RUNES workshop**, Marseille, France, 6 October
Title: Linear stability and attraction basins of oscillators with group interactions

CENTURI Scientific Day, Marseille, France, 17 September
Title: Cooperation by non-kin during birth underpins sperm whale social complexity

Perspectives in Nonlinear Dynamics 2025, Lucca, Italy, 23 July
Title: Sync with higher-order Interactions: Effects on basins and linear stability

Branching networks colloquium: From experiment to theory, Paris, France, 27 June
Title: Growth and structure of underground fungal networks

HONS+AI satellite @NetSci (keynote), Maastricht, Netherlands, 2 June
Title: Higher-order interactions: dynamics, reduction, and brain dynamics

Physical Networks satellite @NetSci, Maastricht, Netherlands, 2 June
Title: Growth and structure of underground fungal networks

Dynamics of Higher-order Interaction Networks @SIAM DS, Denver, USA, 14 May
Title: Sync with Higher-Order Interactions: Effects on Basins and Linear Stability

Seminar @Santa Fe Institute, Santa Fe, USA, 5 May
Title: Synchrony, Whales, and Underground Fungi: Emergence from Local Rules

2024 **AATRN Networks Seminar**, Online, 3 December

Software and Data for Complex Systems satellite @CCS, Exeter, UK, 3 September
Title: Revealing patterns in brain activity with persistent homology

2022 **MB2022**, IPAM, Los Angeles, USA, 31 August
Title: XGI, compleX Group Interactions

Biomat 2022, Granada, Spain, 18 November

2021 **XIX Curso Boliviano de sistemas complejos**, Online, 5 October

Contributed talks (28)

2026 **IV Simposio Internacional de Mycorrhizal Symbiosis in South America**, Quito, Ecuador, 26 August

2026 **Dynamics Days Europe**, Lisbon, Portugal, 23 July

2025 **NetSci 25**, Maastricht, The Netherlands, 4 June

COMPLEX School Annual Meeting, Louvain-la-Neuve, Belgium, 24 April

2024 **Conference on Complex Systems CCS2024**, Exeter, UK, 5 September

NetSci-X, Venice, Italy, 23 January

2023 **Dynamics Days Europe**, Naples, Italy, 12 September

OHBM, Montreal, Canada, 25 July

NetSci, Vienna, Austria, 12 July

2022 **Conference on Complex Systems CCS2022**, Palma de Mallorca, Spain, 21 October

Conference on Complex Systems CCS2022, Palma de Mallorca, Spain, 18 October

2021 **BeNet2021**, Namur, Belgium, 18 November

Networks 2021, Online, 5-10 July

NetBioMed (Networks2021 satellite), Online, 25 June

Complenet Live, Online, 24-26 May

2020 **CENTURI day, Aix-Marseille University**, Marseille, France, 20 November

Belgian Network Research Meeting (BeNet), Brussels, Belgium, 12 November

Toponets (NetSci satellite), Rome, Italty, 18–19 September

Networks and Molecular Biology winter school, Marseille, France, 2–6 March

2019 **Belgian Network Research Meeting (BeNet)**, Hasselt, Belgium, 22 February

2018 **COSMOS meeting**, Dolenjske Toplice, Slovenia, 24–27 September

Workshop on Long-range Interactions and Synchronization, São Paulo, Brazil, 28–31 June

2017 **COSMOS toolbox laboratory**, Brijuni, Croatia, 8–13 October

Dynamics Days, Szeged, Hungary, 5–9 June

COSMOS retreat, Wittenberg, Germany, 26–31 March

2016 **COSMOS workshop 2**, Amsterdam, Netherlands, 11–16 December

2016 **COSMOS school 2**, Aberdeen, UK, 27 June–6 July

2013 **Quantum Physics and Logic (QPL) workshop**, Barcelona, Spain, 17–19 July

Seminars

2024 **naXys, Université de Namur**, Namur, Belgium, 21 November

2021 **naXys, Université de Namur**, Namur, Belgium, 18 February

2021 **CENTURI seminar, Aix-Marseille University**, Marseille, France, 4 February

2019 **naXys, Université de Namur**, Namur, Belgium, 16 May

2019 **Department of Network and Data Science, Central European University**, Budapest, Hungary, 29 March

2018 **Université Libre de Bruxelles**, Brussels, Belgium, 3 April

2017 **University of Florence**, Florence, Italy, 7 December

Posters

2022 **Oustanding challenges in nonlinear dynamics**, Les Houches, France, 21–25 March

2020 **NetSci**, Rome, Italy, 17–25 September

NetBioMed2020 (Netsci satellite), Rome, Italty, 17 September

Alea day, Marseille, France, 7 February

2019 **4th CENTURI day**, Marseille, France, 22 February

IBDM days, Arles, France, 3–4 October

Workshop on Higher-order Interaction Networks: Dynamics, Structure, Data, Oxford, UK, 9–11 September

Complenet, Tarragona, Spain, 18–21 March

2018 **Analysis and Modeling of Complex Oscillatory Systems (AMCOS)**, Barcelona, Spain, 19–23 March

2017 **STATPHYS26**, Lyon, France, 18–22 July

2016 **International conference on biological oscillations (ESGCO)**, Lancaster, UK, 10–14 April

2015 **Lancaster University Christmas Conference**, Lancaster, UK, 15 December

Organisation and service

Organisation

2022– **"Toponets" satellite meeting at NetSci 22, 23, 24, 25, 26**, *Co-Organiser*

2025 **"Structure and Dynamics of Higher-order Systems" minisymposium at SIAM DS**, *Co-Organiser*

2023 **"Tutorial Thursdays"**, *Organiser*, monthly internal tutorials at CENTAI Institute

2018 **"Analysis and Modeling of Complex Oscillatory Systems" (AMCOS) conference**, *Co-Organiser*, Barcelona, Spain, 100+ participants

Reviewer

Physical Review Letters • Nature Communications • Nature Methods • Philosophical Transactions of the Royal Society A • Journal of the Royal Society Interface • Physical Review Research • Chaos • Physical Review E • npj Complexity • Journal of Physics: Complexity • Chaos, Solitons & Fractals • Communications Physics • SIAM Journal on Applied Dynamical Systems • PLOS ONE • Scientific Reports • Fluctuation and Noise Letters • Communications in Nonlinear Science and Numerical Simulation • Springer book: Physics of Biological Oscillations

Program committee member

CompleNet 26 • Complex Networks 23,25 • CCS 24 • FRCCS 22-23

Representation and memberships

2024- **UNamur's Maths Department council**, *Non-tenured scientists representative*

2022-2025 **Council of the Complex Systems Society**, *Elected member*

2015-2018 **Supervisory board of the COSMOS Marie Curie ITN**, *Early Stage Researchers representative*

2011-2012 **Physics Department**, *Student representative*, Université Libre de Bruxelles, Brussels, Belgium

Teaching and outreach

Mentoring

'*' indicates that I was the main advisor.

2025- **Quentin Petitjean***, *MSc Ecological modelling*, University of Rennes

2025- **Renée Locreille***, *PhD candidate*, UCLouvain, co-supervised with S. Declerck
Thesis: "Analyse comparative des réseaux mycéliens des champignons mycorhiziens à arbuscules (CMA) pour la sélection et la gestion de souches adaptées à la production agricole durable"

2024-2025 **Renée Locreille***, *MSc Bioengineering*, UCLouvain, co-supervised with S. Declerck
Master's thesis: "Caractérisation du réseau et de la croissance du mycélium extraracinaire des champignons mycorhiziens à arbuscules (CMA)"

2024-2025 **Gianmatteo Ciulli**, *MSc Physics*, University of Turin, co-supervised with G. Petri

2023-2024 **Noemi Aime***, *BSc Physics*, University of Turin, co-supervised with G. Petri
Title: On the state-nonstate theory of hypnosis: Network and topological EEG findings

Summer 2023 **Francesco Guadagnolo**, *BSc Maths*, University of Turin, co-supervised with G. Petri

Oct 21-Sept 23 **Antonio Leitao**, *pre-doctoral student*, CENTAI Institute, co-supervised with G. Petri

Summer 2020 **Alex Townsend-Teague***, *Undergrad*, University of Oxford
Title: Inferring cell cycle phases from a partially temporal network of protein interactions

Summer 2020 **Arthur Morris***, *Undergrad*, University of Oxford
Title: Inferring cell cycle phases from a partially temporal network of protein interactions

Teaching

Fall 24 & 25 **Dynamical systems**, *MSc in Mathematics*, University of Namur, Belgium

Spring 23 & 24 **Complexity in social systems**, *MSc Physics of Complex Systems*, University of Turin, Italy

Nov 2022 **Synchronisation with higher-order interactions**, *Invited Lecturer*, BIOMAT summer school, Granada, Spain

Oct 2021 **Bifurcations**, *Invited Lecturer*, XIX Bolivian course on complex systems

Spring 2021 **Introduction to Artificial Intelligence (S8BIO2021)**, *Bioingeneering*, École Centrale Marseille, France

2010–2015 **Private tuitions in physics and maths**

2014 **Teacher at Reussit'school**, *individual tuitions*

Outreach

2025 **Dissemination event "La Complessità è Semplice?" organised by "La Scienza Coatta"**, *Siena, Italy*, 3 Sept

2025 **NetPlace panelist: "Science as a craft: Balancing career and skill progression"**, *Maastricht, The Netherlands*, 2 June

2025 **Rallye Mathématique**, *UNamur, Belgium*, 28 March

2018 **Interview and videos by the STA (Slovenian Press agency)**, *Novo Mesto, Slovenia*, 15 September

2017 **Interview for Radio Moka**, *Florence, Italy*, 17 November

2017 **Masterclass "Waves and oscillations"**, *Lancaster University, UK*, 13 July, We hosted 30 A-level students within the programme Headstart from all over the UK

2017 **Masterclass "Waves and oscillations"**, *Lancaster University, UK*, 1 March, We hosted 9 A-level students

Awards and grants

2025 **Best presentation**, *Complex annual meeting*

2020 **Travel grant**, *from FNRS, "Scientific Stay Grant OUT"*

2023 **Nominee for the Thom Young Researcher Award**, *FRCCS*

2023 **Postdoctoral fellowship**, *from FRNS (Belgium)*

2020 **Travel grant**, *from COSTNET, Short Term Scientific Mission (STSM)*

2019 **Travel grant**, *from COSTNET, Short Term Scientific Mission (STSM)*

2018 **Best group project "Plan Bee"**, *at the Mediterranean School of Complex Networks (MSCx)*, Salina, Italy

2015 **COSMOS PhD Fellowship**, *Marie Curie ITN EJD*

2011 **Gold medal with the ULB-Brussels Team**, *at the international Genetically Engineerd Machine competition (iGEM) organised by MIT*, Amsterdam, Netherlands

Computer skills and projects

Languages Python, Matlab, Fortran, Java, Prolog, LaTeX

OS Linux, macOS, Windows

Code **CompleX Group Interactions (XGI):** python package that provides data structures and algorithms for modeling and analyzing complex systems with group (higher-order) interactions,
`xgi.readthedocs.io/`

Hypersync: python code to simulate and analyze synchronization on higher-order networks,
`https://github.com/maximelucas/hypersync`

Phasik: python package to infer temporal phases in temporal networks,
`phasik.readthedocs.io/`

AMCOS_booklet : a LaTeX template for conference booklets,
`github.com/maximelucas/AMCOS_booklet`

Languages

- French: mother tongue
- English: fluent (IELTS: 8/9 in 2015)
- Italian: advanced
- Dutch: advanced
- Hungarian: beginner

Interests

Music (play the violin), football, squash, hiking, former youth movements leader